Chapter 17

Employee Stock Options and Equity Compensation

Every option in this book so far has been a traded contract. Someone wrote it, someone bought it, and either could close the position at any time by trading it back. Chapters 13 through 16 rest on that: a price is the cost of a replicating strategy, and the replicating strategy requires that positions can be established and unwound.

None of this holds for an employee stock option. It cannot be sold. It is forfeited on departure. Its holder is unhedged, undiversified, and forbidden from shorting the stock. It runs ten years when listed stock options run at most about three. And it is issued by the company on its own shares, so exercise creates stock rather than transferring it.

The result is an option whose value depends on human behavior, and the models in this chapter say so explicitly. That is not a retreat from Chapters 13–16. It is those chapters applied consistently: when replication is impossible, the price is no longer pinned by arbitrage, and what fills the gap must be stated as an assumption rather than left implicit.

The chapter also covers the instruments that have largely displaced options in listed-company pay, restricted stock units and performance shares, and the accounting rules that drove the change. Section 10.10 promised this treatment for both warrants and employee options, and the two are the same problem: an option written by the issuer on its own equity.

Sections 17.1 and 17.2 are the contract and why it is not a traded call. Sections 17.3 to 17.5 value it: the models, the dilution that exercise causes, and the exercise decision itself. Section 17.6 is the accounting, Section 17.7 the restricted stock units and performance shares that displaced options, and Section 17.8 private-company equity. Sections 17.9 and 17.10 are warrants and the backdating episode, and Section 17.11 who uses equity pay and why.

17.1 THE CONTRACT

A grant of employee stock options specifies more terms than a listed contract does.

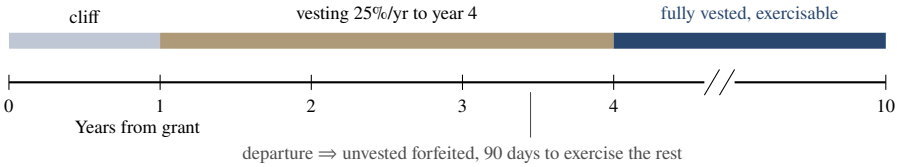


Figure 17.1: A typical grant: one-year cliff, then vesting to year four, ten-year term. The axis is broken between years 4 and 10.

Vesting. The option cannot be exercised until it vests. The common US pattern is a one-year **cliff** (nothing vests until the first anniversary, then 25% at once) followed by monthly or quarterly vesting to year four. Figure 17.1 shows the shape. Unvested options are forfeited outright on departure, which makes vesting the retention mechanism the grant exists for.

Term. Ten years from grant is standard, though four-year and seven-year terms appear. Compare this with the roughly three-year outer limit on single-stock LEAPS from Section 10.10: an employee option outlasts any standard listed option on a stock.

The post-termination exercise window. On leaving, an employee typically has 90 days to exercise vested options or lose them. This clause does most of the damage to the value, because it turns a ten-year option into one that expires 90 days after an event neither party controls.

Non-transferability. The option cannot be sold, pledged, or given away. There is no secondary market and no way to monetize it except by exercising and selling.

Strike. The strike is almost always the market price on the grant date. Sections 17.6 and 17.10 explain why: an at-the-money grant used to carry no accounting charge at all, and still carries the cleanest tax treatment.

Dilution. Exercise creates new shares. The company receives the strike price in cash and issues stock, so the share count rises. Section 17.4 works out what this does to the valuation.

17.2 WHY AN EMPLOYEE OPTION IS NOT A TRADED CALL

Four of the restrictions above break the arbitrage argument, all in the same direction.

The holder cannot hedge. Chapter 13's replication argument requires that a position be constructed and maintained. An employee cannot short the company's stock (most equity plans forbid it and insider-trading policy forbids the rest), so the option's value to the holder is not the cost of replicating it. It is whatever an undiversified, risk-averse person with a concentrated position thinks it is worth, which is *less*.

The holder is undiversified. An employee's human capital is already invested in the firm. Adding options on the same firm concentrates rather than diversifies, so the marginal option is worth less to the employee than to a diversified investor. The size of the gap depends on how risk-averse the holder is and on how much of their wealth is already tied to the firm, and the utility-based estimates run wide: Hall and Murphy (2002) put an executive's **certainty-equivalent** value (the sure sum of cash the executive would accept in place of the option) at roughly half to two-thirds of the Black–Scholes–Merton cost for a typical at-the-money grant, and Meulbroek (2001) found discounts of the same order for concentrated holdings. Whatever the number, it is a real cost of paying in options, and it appears nowhere in the accounts.

Forfeiture is a random early expiration. Departure forfeits unvested options and leaves 90 days to exercise vested ones. Modeling this requires an assumption about employee turnover, and the assumption matters: Section 17.3 shows a value moving from 16.10 to 12.59 as the assumed exit rate goes from zero to 6% a year.

Early exercise is the norm, not the exception. Section 11.8 proved that an American call on a non-dividend-paying stock should never be exercised early, and employees exercise early constantly. Bettis, Bizjak and Lemmon (2005) find ten-year options exercised a median of 4.25 years before expiration, under six years into the term; Carpenter's (1998) sample of top executives gives a mean of 5.83 years after grant, at a stock price 2.75 times the strike. Five to six years at two to three times the strike is the working summary, and it is where the exercise multiple of Section 17.3 comes from.

This is not employees being irrational. The proof in Section 11.8 assumes the holder can sell the option instead. An employee cannot. For a holder choosing between exercising and holding an unsellable, unhedgeable, forfeitable claim on a single company, exercising to diversify is the sensible choice. The theorem does not fail; its hypothesis does.

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That said, exercise-and-hold is common and usually a mistake: it gives up the remaining time value and, in the US, prepays ordinary income tax on the spread to acquire exposure the option already provided (Example 17.4).

17.3 VALUING AN EMPLOYEE OPTION

Three approaches are in use. They disagree, and by how much is the main finding.

1. Black–Scholes with the contractual term. Put the ten-year term into the formula of Chapter 15 and read off the answer. This overstates the value, because it ignores forfeiture and early exercise entirely.

2. Black–Scholes with an expected term. Replace the contractual term with the expected time to exercise or forfeiture, estimated from the company's own history. This is the workhorse: it is what most disclosures use, it is auditable, and it is defensible precisely because it makes no claim to be a model of behavior. A company whose own history gives no reasonable basis for the estimate may use the simplified method of the SEC's Staff Accounting Bulletins (SABs) 107 and 110, which sets the expected term of a plain at-the-money grant midway between vesting and the contractual term: 6.5 years for Example 17.1's grant.

Its weakness is that expected life is not a parameter of the option but an *outcome*, and it depends on the stock price path. A company whose stock triples will see exercises much earlier than one whose stock stagnates, so the expected term estimated from history is conditioned on a history that will not repeat.

3. A lattice with an exercise rule. Build the tree of Chapter 13 and replace optimal exercise with a behavioral rule: the employee exercises voluntarily once the stock reaches a multiple M of the strike, and is forced to exercise (or forfeit, if unvested) on departure, which arrives at rate λ. This is the Hull–White approach, and it is the one that spells out what the other two conceal.

Example 17.1 Three answers for one grant

A company grants at-the-money options: $S_0 = K = 40$, ten-year term, three-year cliff vesting, $\sigma = 30\%$, $r = 4\%$, dividend yield $q = 1\%$.

Method	Value	% of grant price
Black–Scholes, 10-year contractual term	16.704	41.8%
Black–Scholes, 6.5-year simplified term	13.808	34.5%
Black–Scholes, 6-year expected term	13.295	33.2%
Black–Scholes, 4.5-year expected term	11.552	28.9%
Lattice, $M = 2.5$, exit rate 3%/yr	14.22	35.5%

The spread is 16.704 to 11.552, a factor of 1.45. ASC 718 rules out the first row, which ignores early exercise, but the methods it allows still span 15.56 (the lattice's value conditional on vesting, the standards' basis; Example 17.5) to 11.552, and on one million options that vest that is a difference of $4.0 million in reported compensation expense, produced entirely by choices about the model rather than about the contract.

The lattice makes the sensitivity explicit:

	exit 0%	exit 3%	exit 6%
$M = 2.0$	—	13.54	—
$M = 2.5$	16.10	14.22	12.59
$M = 3.0$	—	14.52	—

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Turnover matters more than the exercise multiple: moving the exit rate from 0% to 6% costs 3.51 of value, while moving the exercise multiple from 2.0 to 3.0 gains only 0.98, less of it from 2.5 to 3.0 than from 2.0 to 2.5, because a higher threshold matters only on the paths that reach it.

The practical implication is that the most important input is a human-resources (HR) statistic. The company's own attrition rate drives the number more than volatility does over the plausible range, and a firm with high turnover is genuinely granting less, option for option.

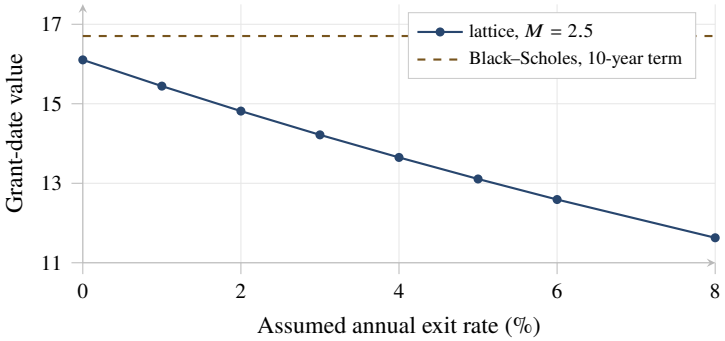


Figure 17.2: Lattice value against the assumed annual exit rate. $S_0 = K = 40$, $\sigma = 30\%$, ten-year term.

Figure 17.2 makes the point visually: the reported expense is close to linear in an assumption that no market quotes and no auditor can verify beyond checking that the company applied its own history consistently.

This is the position of the whole chapter. An employee option cannot be traded, so its reported value is a model output whose most sensitive input is behavioral.

17.4 DILUTION

Exercise creates shares. Existing holders own a smaller fraction of a company that has received the strike price in cash, and **dilution** is the size of that transfer: the fraction of the company passing to the option holders. Three measures of it appear in this section and they are not equal. The *share-count* measure is the new shares as a fraction of the enlarged total, and it is what governs a warrant's value. The *economic* measure is the grant's value as a fraction of market capitalization. The *reported* measure counts only the incremental shares the treasury stock method leaves after assumed buybacks, and Example 17.3 finds it running at about half the economic figure.

Take a company with N shares and no debt, whose assets, excluding anything received from the option holders, will be worth V_T at date T. It has issued M warrants with strike K (in this section M counts warrants; it is no longer the exercise multiple of Section 17.3). If they are exercised, the company receives MK and issues M shares, so the post-exercise share price is

$$\frac{V_T + MK}{N + M}$$

and each warrant is worth that less K. A line of algebra gives

$$\text{warrant payoff} = \frac{N}{N+M}\,\max\!\left(\frac{V_T}{N} - K,\ 0\right) \tag{17.1}$$

So a warrant is $N/(N+M)$ of a call option on V/N, the per-share value of the company *excluding* the warrant proceeds. Both parts of that sentence matter, and the second is where practitioners go wrong. Example 17.2 works through the first.

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Example 17.2 What dilution costs, and what it does not

A company has 10 million shares. It issues 2 million five-year warrants struck at \$40, with $\sigma = 30\%$, $r = 4\%$, $q = 1\%$, and a per-share value of \$40 before the warrants exist.

A five-year call on \$40 struck at \$40 is worth **12.171**. The dilution factor is $10/12 = 0.8333$, so each warrant is worth

$$\frac{10}{12} \times 12.171 = \mathbf{10.143}$$

16.7% less than the equivalent call, which is exactly the fraction of the company the new shares represent.

The total is $2\text{m} \times 10.143 = \$20.3$ million. If the warrants are handed out for nothing, the share price falls by $20.3/10 = \$2.03$ on announcement, from \$40.00 to \$37.97. The value did not vanish; it moved from the existing shareholders to the warrant holders, and the price move is the transfer.

The mistake worth avoiding. Once the warrants or options are outstanding and known, the observed share price already reflects them. Applying (17.1) to the market price then double-counts: the market price is V/N net of the warrant liability, not gross of it.

This has a direct consequence for employee options, and it is the opposite of what intuition suggests. An outstanding employee option grant requires no dilution adjustment at all. The market knows the options exist (they are disclosed), so the share price already embeds them, and the grant-date fair values of Section 17.3 are computed on the observed price with no $N/(N + M)$ factor. The adjustment applies only when valuing an instrument the market has not yet been told about, which for a public company is essentially never.

The dilution that matters for employee options is therefore a governance question about the size of the grant, not a valuation adjustment: how much of the company is being transferred, over what period, and in exchange for what.

Diluted earnings per share

Dilution is quantified and published in only one place, and it ignores the valuation above.

Diluted earnings per share (EPS) is computed by the **treasury stock method**: assume the options are exercised, assume the proceeds are used to buy back shares at the average market price, and count the difference as incremental shares.

Example 17.3 Reported dilution against economic dilution

A company has 10 million shares, net income of \$50 million, and 1 million options struck at \$30. The average share price for the period is \$50. The options were granted at the money with the stock at \$30, worth about \$10 each on the parameters below, and \$6 million of that \$10 million grant-date cost, which is expensed over the vesting period (Section 17.6), remains unrecognized on average over the period.

Basic EPS is $50/10 = \mathbf{\$5.00}$.

Assumed proceeds are $1\text{m} \times 30 = \$30$ million, plus the \$6 million of unrecognized cost (the value of the service employees still owe, which ASC Topic 260 also treats as proceeds): \$36 million, which buys back $36/50 = 0.72$ million shares. Incremental shares are $1.00 - 0.72 = 0.28$ million, so

$$\text{diluted EPS} = \frac{50}{10.28} = \mathbf{\$4.864}, \qquad \text{reported dilution } \frac{0.28}{10} = 2.8\%$$

The reported figure is the increase in the share count. Earnings per share itself falls by 2.7%: a share count 2.8% larger divides into the same earnings.

Valued with the stock at \$50, six years, $\sigma = 30\%$, $r = 4\%$ and $q = 1\%$, each option is worth \$25.68, so the grant is worth \$25.7 million against a market capitalization of \$500 million, or **5.1%**

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of the company.

The reported dilution is roughly half the economic dilution, and the reason is structural: the treasury stock method counts the option at its intrinsic value less unrecognized compensation cost, which ignores time value entirely. It is an accounting convention for allocating this period's earnings, not a measure of what has been given away.

The gap widens exactly where it matters most. For a volatile company whose options are near the money, intrinsic value is small and time value is nearly the whole of the option, so the treasury stock method reports almost no dilution on grants that transfer a substantial fraction of the equity.

An analyst who wants the economic number must compute it, and the inputs are disclosed: option counts, strikes, and grant-date fair values all appear in the equity compensation footnote. Section 17.11 returns to why this matters for reading buyback announcements.

17.5 THE EXERCISE DECISION

Section 17.2 said employees exercise early and that this is not irrational. Example 17.4 works through what the choice actually looks like, because the arithmetic is not obvious and the common advice is wrong in both directions.

Example 17.4 Three things to do with a vested option

An employee holds 10,000 vested non-qualified options (NSOs) struck at \$20: the ordinary kind, taxed as income on the spread at exercise (Section 17.8 covers the incentive kind). The stock is \$50, four years remain, $\sigma = 35\%$, $r = 4\%$, no dividend. Their marginal tax rate is 40%.

The corresponding traded call is worth **33.483** against an intrinsic value of **30.00**, so the time value is **3.483**, or 11.6% of the intrinsic value, and \$34,826 across the position.

Action	Cash today	Position afterward
Hold	\$0	options worth \$334,826 before tax
Exercise and sell	+\$180,000	nothing
Exercise and hold	−\$320,000	\$500,000 of stock

Read the third row carefully. Exercising costs \$200,000 in strike and creates \$300,000 of ordinary income, so \$120,000 of tax comes due whether or not any stock is sold. Holding would defer that tax, not avoid it, so exercise-and-hold prepays it: the employee spends \$320,000 in cash to acquire exposure the option already gave them, and gives up the \$34,826 of time value.

What the prepayment buys is that later appreciation is taxed as a capital gain rather than as ordinary income. That is the real content of "I want to start the capital gains clock," an offset worth weighing on a stock expected to rise far, though the time value, the deferral and the concentration usually outweigh it. "I believe in the company" is no reason at all: belief is an argument for keeping the exposure, and the option *is* the exposure, with leverage and without the cash outlay.

The case for exercising is the case for diversifying, and it is the first two rows that are in genuine tension. They must be compared after tax on both sides, since the options also owe the tax when eventually exercised: at a constant 40% rate the \$334,826 is worth $0.6 \times \$334{,}826 = \$200{,}896$ after tax, against \$180,000 of cash from exercising and selling. Diversifying therefore costs about \$20,900, the after-tax time value. An employee whose net worth is already concentrated in the firm may rationally pay that, and Section 17.2's undiversified-holder discount is exactly the statement that the model value is not what the position is worth to them.

Two things sharpen the decision. *Time value does not disappear as the stock rises*: at \$120 the same option's time value would still be 2.98, barely changed from the 3.483 here. It never falls to zero: for a non-dividend-paying stock it converges to the interest on the deferred strike, $K(1 - e^{-rT})$, which is 2.96 for this option and is Section 11.8's result seen from the other side. What does fall is the time value measured against what is at stake: 3.0% of the intrinsic value rather than 11.6%, so an employee exercising to diversify surrenders proportionally less at the higher

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price. *A near departure forces it*: the 90-day window converts the choice into a deadline, which is why so many exercises cluster around resignations rather than around stock prices.

17.6 ACCOUNTING AND EXPENSING

Until 2006 a US company with a calendar fiscal year could grant at-the-money options and record no expense whatsoever. Accounting Principles Board Opinion 25 (APB 25, 1972) measured the charge as the option's *intrinsic* value at grant, which for an at-the-money option is zero, and Financial Accounting Standard 123 (FAS 123, 1995), though it encouraged expensing fair value, let companies keep that method with footnote disclosure; nearly all did. Companies that would never have paid $14 million in cash bonuses granted one million options worth $14 million and reported nothing.

FAS 123R (issued in December 2004), now ASC 718, and IFRS 2, in force from 2005, ended that. In outline, the current rules are these:

Measurement at grant-date fair value. The charge is the model value on the grant date, computed as in Section 17.3 but conditional on vesting: departures before vesting reduce the number of awards expensed, not the value of each. It is never remeasured for subsequent stock price moves. An option valued at $15 at grant that expires worthless was still a $15 expense, and so was one that ends $200 in the money.

Recognition over the vesting period. The total is spread across the service period, but the pattern depends on whether the grant is treated as one award or several. **Straight-line recognition** treats it as one award and expenses equal amounts each year; **graded recognition** treats each vesting tranche as a separate award expensed over its own vesting period, so the early years carry a share of every tranche. Example 17.5 sets the two side by side.

Equity or liability classification. Everything above assumes settlement in shares. An award settled in cash is a liability. It may be a **stock appreciation right (SAR)**, which pays the spread without the holder paying the strike, or phantom stock, which pays a share's value. A liability is remeasured to fair value every reporting date until it settles, so a cash-settled SAR, unlike the equity-settled option above, rises and falls with the share price for years and ends at whatever the award pays. Companies accept that volatility to pay the award without issuing shares, though the cost to existing shareholders is the same: the cash is theirs too, so settling in it keeps the share count fixed but transfers the same value. SARs are common where local law restricts issuing shares to employees and among private companies with no market to settle into.

Forfeiture. Expense on awards forfeited for failure to vest is reversed. Since ASU 2016-09, ASC 718 lets a company either estimate forfeitures or account for them as they occur; IFRS 2 requires an estimate. A company that estimates uses the exit rate *twice*, for different purposes: once inside the model, after vesting, to shorten the option's life, and again outside it to reduce the number of awards expected to vest.

Pay versus performance. For fiscal years ending on or after December 16, 2022, US proxy statements must also report "compensation actually paid" (Item 402(v) of Regulation S-K, adopted in August 2022), which remeasures equity awards at fair value at each year end and at vesting. It is the remeasurement the expense never makes, published beside it.

Example 17.5 Straight-line against graded recognition

A grant of 100,000 options vests 25% a year over four years. Each annual tranche is a separate option with its own vesting date, so each has its own value (Example 17.1's lattice, with the cliff set at the tranche's vesting date). The values are the expected cost per option granted, including the chance of leaving before vesting:

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Tranche	Options	Value	Cost
1-year cliff	25,000	14.52	$363,000
2-year cliff	25,000	14.37	$359,250
3-year cliff	25,000	14.22	$355,500
4-year cliff	25,000	14.05	$351,250
Total	100,000		**$1,429,000**

Per option granted, longer vesting costs *less*, not more, because vesting later means more exposure to forfeiture before the option becomes exercisable. The effect is small (3.2% from the first tranche to the fourth), but it runs the opposite way from the usual "more time is worth more" intuition: the extra time is time the holder cannot use. The standards measure fair value conditional on vesting instead, and put pre-vesting departures into the number of awards. On that basis each value is divided by the probability of staying until the cliff, $e^{-0.03c}$ for a cliff of c years, and the tranches are worth 14.96, 15.26, 15.56 and 15.85. These rise with the cliff: once pre-vesting departures are set aside, what costs value is early exercise after vesting, voluntary at the multiple or forced by departure, and a later cliff postpones both. The expected totals are the same on either basis.

Now recognize that $1,429,000 over the vesting period:

	Year 1	Year 2	Year 3	Year 4
Straight-line	$357,250	$357,250	$357,250	$357,250
Graded	$748,938	$385,938	$206,312	$87,812

Graded recognition puts 52% of the total cost in the first year against straight-line's 25%. Both are permitted under ASC 718 for awards with only a service condition (vesting on continued employment alone); IFRS 2 requires the graded pattern. The total is identical and the timing is not, so two companies with identical pay can report materially different first-year expense.

The difference washes out in steady state for a company that grants every year, but *not* for one that has just started granting or made a large one-off award, precisely the cases where the number is being looked at.

Repricing and modification. When the stock falls far enough that outstanding options are hopelessly underwater, they stop retaining anyone. Companies respond by repricing (lowering the strike) or by exchanging old options for fewer new ones or for restricted stock.

Accounting treats this as a modification: the incremental charge is the fair value *added* by the change, measured immediately before and after, on top of any unrecognized original cost. So a repricing is not free, and shareholders usually must approve it. The economics are worse than the accounting suggests: an employee who learns that underwater options get reset has learned that the downside of the option was never real, which removes most of the incentive the grant was supposed to create.

Repricing and exchanging are usually compared on cost; Example 17.6 shows that this framing gets the incentive question backward.

Example 17.6 Repricing against a value-for-value exchange

A company granted 500,000 ten-year options struck at $80. The stock is now $30 with six years remaining; volatility is 45%, the rate 4%, and the stock pays no dividend. The existing options are worth $6.5944 each and a new six-year at-the-money option would be worth $14.6527.

Repricing all 500,000 to a $30 strike adds $14.6527 − $6.5944 = $8.0583 of fair value per option. The alternative is a value-for-value exchange: cancel the underwater options and grant fewer new ones at $30, in a ratio chosen so the fair value handed over matches the fair value taken

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back: \$6.5944/\$14.6527 = 0.45005 new per old, or 225,025 in all. The two compare as follows:

	Reprice	Exchange
Options held afterward	500,000	225,025
Fair value before	\$3,297,207	\$3,297,207
Fair value after	\$7,326,327	\$3,297,207
Incremental expense	**\$4,029,120**	**\$0**

The exchange delivers exactly what it took away, so nothing is recognized. That is why boards prefer it, and why proxy advisers (firms advising institutional shareholders how to vote) accept it.

Now look at what each does to the employee's exposure rather than to the reported number. Incentive intensity is the delta of the grant, how many shares of exposure the employee actually holds:

	Delta	Shares of exposure
Underwater \$80 options	0.452	225,931
Repriced \$30 options	0.779	389,507
Exchanged \$30 options	0.779	175,297

The exchange that costs nothing to report leaves the employee with 22% *less* exposure to the share price than the underwater options they surrendered. A deeply out-of-the-money option has a low value and a surprisingly high delta (nearly half a share), so trading it for a smaller number of at-the-money options is a bad trade in delta terms even when it is a fair trade in value terms.

So the two alternatives fail in opposite directions. The repricing multiplies the value of the employee's grant by 2.2 and their share exposure by 1.7, and charges shareholders \$4.0 million for doing it, having taught everyone that strikes follow the stock down. The exchange reports nothing, costs nothing, and quietly reduces the incentive it was adopted to restore: the one that is easy to disclose does less of what the grant was for.

17.7 RESTRICTED STOCK UNITS AND PERFORMANCE SHARES

Options have been displaced. Most large listed companies now grant the majority of equity compensation as restricted stock units (RSUs), and the reason is visible in one calculation.

An **RSU** is a promise to deliver one share on vesting. Having no strike, it is not an option but a prepaid forward on the stock (equivalently, a zero-strike call) with delivery contingent on service. Its grant-date fair value is the share price, less the present value of dividends forgone during vesting if the units carry no **dividend equivalents**: cash or extra units matching the dividends paid on the shares while the units are unvested.

Example 17.7 Why RSUs displaced options

Take the same company and grant terms: $S_0 = 40$, $\sigma = 30\%$, $q = 1\%$, Example 17.1's three-year cliff and 3% annual exit rate.

The two must be valued on one basis. The option's 14.22 is an expected cost per option granted and carries the chance of leaving before the cliff (Example 17.5), so the RSU must carry it too: a unit without dividend equivalents is a prepaid forward on the share that is forfeited on departure before vesting, worth $40e^{-(0.01+0.03)\times 3} = \mathbf{35.48}$. So one RSU costs the same as **2.50 options** (a ratio unchanged conditional on vesting, where both values rise by $e^{0.09}$), which is why a company moving from options to RSUs grants one unit in place of two to three options.

Now suppose the stock falls immediately after the grant, with the same strike and the same vesting still ahead:

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Stock price	RSU value	change	Option value	change
40	35.48	—	14.22	—
30	26.61	−25%	8.41	−41%
20	17.74	−50%	3.68	−74%

A halving of the stock destroys 74% of the option grant's value and exactly 50% of the RSU grant's. Figure 17.3 shows why: the RSU is a straight line (a prepaid forward is linear in the share price), while the option curve bends away from it and flattens toward zero, so the same fall in the stock takes proportionally more of the option. After 2008 and again after 2022, boards that had granted options found their entire retention apparatus nearly worthless at exactly the moment they most needed to retain people, while boards that had granted RSUs found theirs merely diminished.

That is the whole argument, and it is an argument about the *purpose* of the grant. An option pays only for outperformance; an RSU is deferred stock that pays for staying. If the objective is retention, as it is for most employees below the executive level, the RSU does the job with less accounting expense per unit of retention delivered.

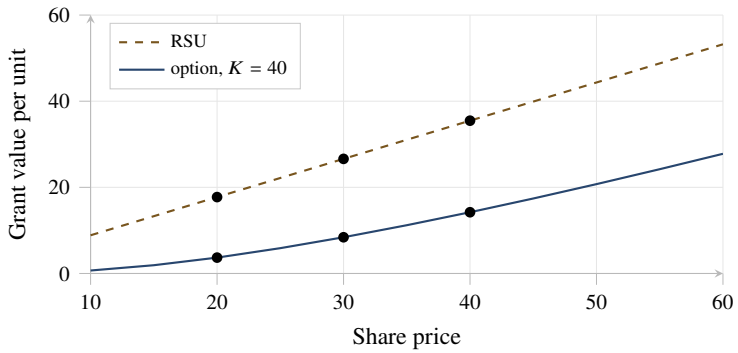


Figure 17.3: Grant value against the share price, on Example 17.7's terms. Marks show the table's three cases.

Performance share units. At executive level, grants increasingly vest on a performance condition rather than on service alone, and the accounting treats two kinds differently.

A **non-market condition** (revenue, earnings per share, a regulatory approval) is not built into the fair value. The grant-date value is computed as if the condition will be met, and the number of awards expected to vest is estimated and trued up each period. If the target is missed, the expense is reversed entirely.

A **market condition**, such as total shareholder return (TSR: price appreciation plus dividends) measured against an index or a peer group, *is* built into the fair value and is never trued up. The award is valued at grant with a Monte Carlo simulation of the company's stock against the benchmark, and that value is expensed whether or not the target is hit, provided the employee completes the service period.

The asymmetry is easily missed. A relative TSR award that pays nothing still generates expense for its full grant-date value, because the possibility of paying was what was purchased. An EPS-target award that pays nothing generates no expense at all. Two grants that look identical in the proxy statement can behave completely differently in the income statement, and Chapter 20's simulation machinery is what values the first of them.

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17.8 PRIVATE COMPANIES AND STARTUP EQUITY

Everything above assumes an observable share price. A private company has none, with consequences beyond valuation.

The 409A valuation. Section 409A of the US Internal Revenue Code requires that options be struck at or above fair market value, and for a private company that value is in practice established by an independent appraisal (a **409A valuation**, which the tax authorities presume reasonable) refreshed at least annually or after any material event. If the strike turns out to be below fair market value, the option fails Section 409A, and the statute puts the tax on the employee, not the company: the in-the-money amount is taxed as the option vests, plus a 20% additional tax and interest, on a gain the employee cannot sell. The employee cannot check the valuation, which is why companies buy the appraisal (the tax authorities can overturn it only by showing it grossly unreasonable) and why some reimburse employees when one fails.

The appraisal is a genuine exercise in valuation under uncertainty: comparable-company multiples, discounted cash flow, and an allocation of the resulting equity value across the preferred and common share classes, since the common stock employees receive sits behind the preferred holders' **liquidation preferences**, their right to be paid back first when the company is sold or wound up. That subordination is why the 409A price routinely sits far below the price paid in the latest preferred round (often under half at an early-stage company, a gap that narrows only as an exit approaches), and why an employee's strike price can be far below the company's headline valuation.

Illiquidity has no offsetting benefit. The employee holds an option on an asset with no market, so the diversification argument of Section 17.2 bites hardest here. There is not even the escape of exercising and selling; Example 17.8 shows the cost to a leaver.

Example 17.8 The exercise-and-hold trap

An engineer holds 100,000 vested incentive stock options (ISOs) struck at \$0.50. The latest 409A valuation is \$8.00. They leave the company, and the 90-day window starts.

Exercising costs $100{,}000 \times 0.50 = \$50{,}000$ in cash. That is the visible cost. The invisible one is the tax. Exercising an ISO creates no regular taxable income, but the alternative minimum tax (AMT), a parallel US tax computation, counts the spread as income:

$$(8.00 - 0.50) \times 100{,}000 = \$750{,}000$$

which at the 28% AMT rate produces a bill on the order of **\$210,000** the following April, on an unrealized gain in shares that cannot be sold.

The total cash required is roughly \$260,000, against a position that may be worth nothing. The alternative is to let the options lapse and walk away from a nominal \$750,000.

This is not a rare edge case; it is the standard situation for anyone leaving a private company with appreciated options, and it is why several structural responses exist: extended post-termination windows (some companies now allow seven or ten years, though an option exercised more than three months after departure loses ISO status), company-run tender offers that create periodic liquidity, and third-party financing against the shares. Each of these is a response to the same defect: an instrument that generates a tax liability before it generates cash.

Incentive against non-qualified options. Incentive stock options avoid ordinary income tax at exercise but trigger AMT as above, and keep the favorable treatment only if the shares are held two years from grant and one from exercise; at most \$100,000 of stock, valued at grant, may first become exercisable as ISOs in a year. Non-qualified options are taxed as ordinary income on the spread at exercise, withheld through payroll, with no AMT complication, and the company deducts that income; an ISO gives it no deduction unless the shares are sold early. The choice turns on whether the employee will hold long enough to qualify.

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Why startups grant options at all. Cash-constrained companies pay in equity because they have equity and not cash, and the option's leverage means a given accounting expense buys a larger notional bet on the company's success, which is precisely the alignment a startup wants and a mature company does not. The instrument matches the situation. It is when a mature, cash-generating company keeps using it out of habit that the mismatch of Section 17.7 appears.

17.9 WARRANTS BEYOND EMPLOYEE OPTIONS

Section 10.10 introduced warrants alongside employee options. Three families are worth distinguishing, because only two of them dilute.

Corporate warrants are issued by the company, usually attached to a bond or preferred issue as a sweetener, and exercise creates shares. Equation (17.1) applies. The attachment is the point: a warrant lets an issuer sell debt at a lower coupon by handing the buyer equity upside, the trade of Chapter 25's convertible bond with the two parts separable.

Special purpose acquisition company (SPAC) warrants were the highest-volume example of the 2020–21 boom. A SPAC is a listed shell company that raises cash from public investors to merge with a private operating company, which thereby becomes listed. It sells units (typically one share plus a fraction of a warrant), with the warrant struck around 15% above the unit price and running five years from the merger. They dilute, they are usually redeemable by the issuer once the stock trades well above the strike, and that redemption right, a call the issuer holds on the warrant, caps the holder's upside. Redemption and, before the merger, the risk that no deal closes and the warrants expire worthless are why SPAC warrants trade well below the plain Black–Scholes–Merton value of their terms, and a valuation that ignores them is badly wrong.

Covered warrants are issued by a bank, not by the company, on someone else's stock. They are the dominant retail option instrument in Hong Kong, Germany, and several Asian markets, and they are simply calls and puts in a retail wrapper: exercise settles in cash or in existing shares, so there is no dilution and (17.1) does not apply. What they carry instead is issuer credit risk (the uncollateralized counterparty exposure that Chapter 9's CVA prices) and a bid–ask spread set by a single market maker who is also the issuer.

The test is always the same question: does exercise create a share or move one? If it creates one, the dilution factor applies and the instrument is a claim on the company. If it moves one, the instrument is an ordinary option and the issuer is just a counterparty.

17.10 BACKDATING

Between roughly 1996 and 2005, a substantial number of US companies granted options and then chose the grant date afterward, selecting a day on which the stock had been low, and reported the grant as at-the-money. It was not.

The mechanism was pure accounting arbitrage. Under APB 25 an at-the-money grant carried zero expense and an in-the-money grant carried an expense equal to its intrinsic value. Backdating converted the second into the first in the records while delivering the second in substance. Example 17.9 puts numbers on both halves of that statement.

Example 17.9 The arithmetic of choosing the lowest price

Take a stock with 45% annual volatility, $r = 4\%$ and $q = 1\%$. Over a 21-trading-day month, the expected minimum is **8.3%** below the expected price on a randomly chosen day.

Grant 500,000 ten-year options honestly at the month's typical price of \$40 and the strike is \$40. Backdate to the low and the strike is \$36.67.

The option value rises from 21.478 to 22.190, only 3.3%, or \$356,000 across the grant, because a ten-year option at 45% volatility is already worth 54% of the stock price and an 8% strike

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reduction moves it little.

The intrinsic value is the real number. The backdated option is immediately $3.33 in the money, so under APB 25 it carried a required expense of

$$3.33 \times 500{,}000 = \$1.67 \text{ million}$$

which was reported as zero.

That gap is the point: little value to the executive, a great deal of concealed expense to the company. It was the misreporting, not the transfer, that made it a securities-law violation rather than a compensation decision.

Case Study 17.1 How backdating was found, and what ended it

The practice was uncovered statistically rather than by a whistleblower. Erik Lie, an academic studying option grant dates, noticed that stock returns were systematically *negative* before grant dates and *positive* after, across hundreds of companies, over years.

No plausible innocent story fits that pattern. Executives cannot reliably forecast a bottom, and boards do not schedule compensation meetings for market troughs. A pattern of buying at the low across a large sample is neither skill nor luck. It is hindsight.

What ended it was not enforcement but a change in reporting infrastructure. The Sarbanes–Oxley Act of 2002 required insiders to report option grants within **two business days** rather than, as previously, up to 45 days after the end of the fiscal year. Backdating requires a reporting lag, and the lag was removed. Heron and Lie (2007) show the anomaly shrinking sharply after 2002 and surviving mainly among grants reported late.

The episode matters for a reason beyond the fraud: it is a clean case of a disclosure rule making a behavior mechanically difficult rather than merely illegal, as Section 2.7's position limits and Section 2.11's clearing mandate did.

By March 2007 the research firm Glass Lewis counted 257 companies with internal reviews or SEC or Justice Department investigations, the SEC staff put its own count at about 140, dozens restated earnings, and a small number of executives were prosecuted. The much larger effect was on practice: grants are now made on pre-scheduled dates by formula, a lasting improvement bought with a scandal. A restatement now carries a further cost: under the SEC's Rule 10D-1, adopted in 2022 and in force through the exchanges' listing standards since October 2, 2023, a listed company must recover, regardless of fault, the incentive pay its executive officers received in the three fiscal years before a required restatement in excess of what the restated figures would have paid.

Grant timing today. The successor to backdating is **spring-loading**, granting just before good news is released. SAB 120 (November 2021) says such an award's fair value should reflect the pending news, and Item 402(x), adopted in December 2022, requires a company to state its grant-timing policy and tabulate executive option grants made from four business days before to one business day after a 10-Q, a 10-K or an 8-K disclosing material nonpublic information. The same release imposed a 90- to 120-day cooling-off period on officers' and directors' Rule 10b5-1 trading plans, which carry much of the selling of Section 17.11.

17.11 WHO USES EQUITY COMPENSATION, AND WHAT FOR

The instruments in this chapter exist to solve specific problems, and each solves a different one.

Retention is the dominant purpose below executive level, and RSUs do it best. Unvested units are forfeited on departure, so the employee carries a rolling balance of value they lose by leaving. An option grant achieves the same thing only while it is above water, which is why the

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drawdown behavior shown in Example 17.7 drove the shift to RSUs.

Alignment is the argument for options and for performance shares. An option pays only if the share price rises above the grant-date level, so the holder's payoff resembles a shareholder's, levered. Whether this genuinely changes behavior is contested; what is not contested is that it changes the *distribution* of the executive's payoff, and a convex payoff rewards volatility. An executive paid in options has a personal interest in the firm taking more risk, aligned with shareholders only when they want more risk, and never with creditors (Chapter 24).

Cash conservation is the startup's reason (Section 17.8).

Tax and accounting arbitrage was a reason and is much less of one. From 1994, Section 162(m) of the Internal Revenue Code capped a public company's deduction for each top executive's pay at $1 million but exempted performance-based pay, at-the-money options included, a push toward options that the 2017 tax act removed. The zero-expense era ended in 2006 and the reporting lag closed in 2002. What remains is genuine but modest: the corporate deduction on option exercise can exceed the recorded expense when the stock has risen, producing a windfall tax benefit.

Where this shows up in market prices. Two channels connect the chapter to the rest of the book. Large option and RSU programs generate predictable, calendar-driven *selling* (vesting dates and same-day-sale exercises, in which the shares are sold at once to cover the strike and the tax), which single-stock market makers track explicitly. And companies that grant heavily buy back stock to offset dilution, making buyback programs partly a mechanical consequence of the compensation plan rather than a signal about valuation. Reading a buyback announcement as a statement about intrinsic value, when it is offsetting a vesting schedule, is the same category of error as reading a borrow cost as a directional view.

SUMMARY

An employee stock option is non-transferable, forfeited on departure, exercisable only after vesting, and long-dated. Those restrictions break the replication argument on which Chapters 13–16 rest, so its value is not enforced by arbitrage and must be modeled with a behavioral assumption stated explicitly. Employees exercise early (typically about halfway through the term, once the stock has risen far above the strike), which does not contradict Section 11.8's theorem: its hypothesis, that the holder can sell the option instead, does not hold. Defensible methods give materially different values for the same grant, and the most sensitive input is turnover: an HR statistic, not a market observable.

Dilution is measured three ways and they disagree. An outstanding, disclosed grant needs no valuation adjustment at all, because the share price already reflects it, and applying the $N/(N+M)$ factor to the market price double-counts. The reported measure, the treasury stock method behind diluted EPS, understates economic dilution, and the gap is widest exactly where time value dominates: near-the-money options on volatile stocks.

For the holder, exercising and holding is usually the mistake: it spends the strike, prepays the tax and gives up the time value to acquire exposure the option already provided, in exchange for capital-gains treatment of later appreciation. The case for exercising is diversification, not belief in the company, and its price is the after-tax time value; belief in the company is an argument for keeping the option.

Expensing at grant-date fair value replaced intrinsic-value accounting in 2005–06, and the value is never remeasured, so an option that expires worthless was still an expense. Graded recognition moves the cost well forward of where straight-line puts it. Performance awards divide on the same principle: a market condition is priced into the grant-date value and never trued up, while a non-market condition is trued up and reversed if missed. RSUs have displaced

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options at most large listed companies, and the reason is drawdown behavior: a fall in the stock destroys a larger fraction of an option grant's value than of an RSU's, which loses value exactly in proportion to the price. Repricing underwater options is expensed as the value it adds; a value-for-value exchange reports nothing but leaves the employee with less exposure to the share price than the options surrendered, because a deep out-of-the-money option carries a high delta for its value.

Private companies strike at a 409A value below the preferred round, because common stock is subordinate, and leaving with appreciated ISOs creates an AMT liability on a gain that cannot be sold to pay it. Warrants divide by whether exercise creates a share or moves one: corporate and SPAC warrants dilute, while a covered warrant issued by a bank on someone else's stock is an ordinary option carrying issuer credit risk. Backdating exploited the old intrinsic-value rule to conceal a substantial expense. It was detected statistically and ended by Sarbanes–Oxley's two-day reporting rule, a disclosure change that made the practice mechanically difficult rather than merely illegal.

FURTHER READING

Hull and White's (2004) paper on employee stock option valuation sets out the exercise-multiple lattice of Section 17.3 and is the direct source for the approach, with Huddart and Lang (1996), Carpenter (1998) and Bettis, Bizjak and Lemmon (2005) supplying the exercise evidence it is calibrated to; it is also explicit about what the model does and does not claim. Hall and Murphy (2002) and Meulbroek (2001) are where the undiversified-holder discounts quoted in Section 17.2 come from; both make plain how much the answer depends on the risk aversion assumed. Lie (2005) and Heron and Lie (2007) are the empirical backdating papers and repay reading as an exercise in inference from a distribution rather than from a document. On the accounting, ASC 718 and IFRS 2 differ in ways that matter for the graded-recognition question of Example 17.5, and any comparison of US and European compensation disclosures should start there. The practitioner literature on 409A valuation is largely produced by appraisal firms and should be read with that in mind.

PRACTICE QUESTIONS

17.1 Explain why Section 11.8's theorem on early exercise does not apply to an employee option, being precise about which assumption fails.

17.2 A grant has a ten-year term but an expected life of five years. Explain why using the expected life in Black–Scholes is an approximation rather than a correction, and whether its error has a definite sign.

17.3 Using Example 17.1, state which input the reported expense is most sensitive to, and explain why that is awkward for an auditor.

17.4 A company with 20 million shares issues 5 million warrants, and the equivalent call is worth \$8.00. What is each warrant worth? Explain why an outstanding, disclosed employee grant instead requires no dilution adjustment, and state which of a covered and a corporate warrant needs the factor in (17.1).

17.5 A grant of 200,000 options has a grant-date fair value of \$12 each and vests 25% a year over four years. Compute the first-year expense under straight-line recognition, and compare it with graded recognition.

17.6 Explain why one RSU is worth two to three options, why the ratio would be smaller for a more volatile company, and what happens to the expense of a performance award with a market and with a non-market condition if the target is missed.

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17.7 An employee leaves a private company holding 50,000 ISOs struck at $1.00 with a 409A value of $6.00. Compute the cash needed to exercise and the AMT income created.

17.8 Explain how backdating was detected, and why the Sarbanes–Oxley two-day reporting rule stopped it where prohibition had not.

17.9 A vested option is worth $18.40 with the stock at $60 and a strike of $45. State the time value, explain what an employee gives up by exercising, and use Example 17.4 to explain why "I believe in the company" is an argument against exercising rather than for it.

17.10 A company has 40 million shares, 4 million options struck at $25, and an average share price of $40. Compute the incremental shares under the treasury stock method ignoring unrecognized cost, explain why the method understates economic dilution, and identify the type of company for which the understatement is largest.

17.11 Using Example 17.6, state the incremental expense of the repricing and of the value-for-value exchange, and explain why the exchange is cheaper to report and weaker as an incentive.

17.12 A company grants 100,000 cash-settled SARs, fully vested at grant, worth $8 each; a year later they are worth $13. Explain how the expense differs from an equity-settled option grant of the same value, and what the pay-versus-performance table of Item 402(v) would show for the year.

FURTHER QUESTIONS

17.13 Example 17.1 shows the lattice value falling from 16.10 to 12.59 as the assumed exit rate rises from 0% to 6%. Explain the mechanism by which turnover destroys option value, distinguishing its effect on vested from unvested options, and discuss whether a company with high turnover should regard its option program as cheap or as ineffective.

17.14 Section 17.2 states that an employee values the option below its model value because they are undiversified, and that the gap appears nowhere in the accounts. Construct the argument that the accounting is nonetheless correct, then the argument that it is not, and state what each side takes the expense to be measuring.

17.15 An executive is paid entirely in at-the-money options. Show that their payoff is convex in the share price, and explain what this implies about their preferred level of firm risk. Extend the argument to creditors, and explain why the same grant that aligns the executive with shareholders sets them against bondholders.

17.16 Example 17.7 shows an option grant losing 74% of its value on a halving of the stock while an RSU grant loses 50%. A board argues that this makes options the better instrument because they punish failure. Evaluate the argument, distinguishing the retention objective from the incentive objective, and state what the board's conclusion should be for a senior executive and for a mid-level engineer.

17.17 Example 17.9 finds that backdating raised the option's value by only 3.3% while concealing $1.67 million of expense. Explain why the accounting effect so greatly exceeded the value transferred, and use this to argue that the practice is better understood as a financial-reporting offense than as a compensation abuse. Then explain why the two are hard to separate in a company where executive pay is benchmarked against reported earnings.

17.18 Section 17.11 claims that buyback announcements from heavy option grantors are partly mechanical rather than informational. Design a test that would separate the two motives using publicly available data, stating what you would need and what each hypothesis predicts.

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